

1. Autonomic disturbances: e.g., spontaneous VC or VD, sweating, etc.
2. Body temp. disturbances: Hyper or hypothermia
3. Sleep disturbances:
 - a) Damage of post. hypoth. → Prolonged sleep & narcolepsy
 - b) Irritative lesion of post. hypoth. → Insomnia
4. Disturbed circadian rhythms
5. Endocrinal disorders: e.g., central DI & Fröhlich's syndrome
6. Food intake disturbances: Hyperphagia or anorexia

Lecture (15): Ascending Reticular Activating System, Electroencephalography & Physiology of Sleep

(Dr. Wagdy Nashaat Habeeb)

1. Ascending Reticular Activating System (ARAS or RAS)

➤ Definition:

It is a multi-neuronal polysynaptic system of nerve fibers

Origin: Facilitatory reticular formation (FRF)

Pathway: "Reticulo-thalamo-cortical"

Fibers from facilitatory RF extend upwards → non-specific thalamic nuclei → all areas of CC

N.B. Some fibers project directly to CC without relay in thalamus

➤ Functions:

ARAS stimulates & ↑↑ excitability of cortical neurons →

1. It is responsible for alert response, consciousness & wakefulness
2. It creates optimal cortical excitability for conscious perception of sensory signals

N.B. Its depression → sleep, while its damage, *e.g.*, by vascular lesions & tumors of brainstem → coma

➤ **Factors affecting its activity:**

Factors that ↑↑ ARAS activity	Factors that ↓↓ ARAS activity
1. <u>All sensory signals</u> (especially pain, proprioceptive & auditory signals)	1. <u>Lack of sensory & cortico-fugal signals</u>
2. <u>Cortico-fugal signals</u> , from: (a) Motor cortex: during voluntary movements (b) Limbic association area: during emotions	2. <u>Signals from sleep centers</u> in RF
3. <u>Analeptic drugs</u> , <i>e.g.</i> , CA, amphetamine & caffeine	3. <u>CNS depressant drugs</u> , <i>e.g.</i> , barbiturates & GA (by producing hyperpolarization → preventing the development of EPSPs)

2. Electroencephalography (EEG)

➤ Definition:

It is the record of “spontaneous” electrical activity of the brain

➤ Recording:

Applying electrodes to the scalp of the patient in:

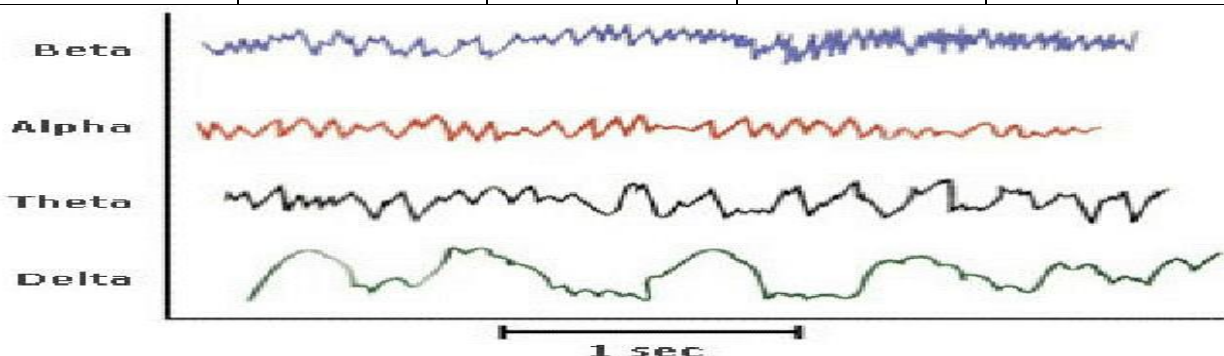
1. Calm room
2. Comfortable temp.
3. Complete physical & mental rest

➤ Waves:

1. Amplitude: 0 – 200 mV
2. Frequency: 1 – 50 Hz

➤ Normal EEG waves:

	Alpha	Beta	Theta	Delta
1. Frequency	8 – 13 Hz	18 – 30 Hz	4 – 7 Hz	1 – 3 Hz
2. Amplitude	50 mV	20 mV	100 mV	100 mV
3. Recorded from	Parietal & occipital	Frontal	Parietal & temporal	----
4. Age & state of person	* <u>Adult:</u> Awaken, relaxed, with closed eyes	* <u>Adult:</u> During brain activity with opened eyes * <u>REM sleep</u>	* <u>Adult:</u> During light sleep * <u>Child</u>	* <u>Adults:</u> During deep sleep & GA * <u>Infant</u>



N.B.: Other waves may be recorded in certain conditions, e.g.:

1. **Gamma waves:** fastest (up to 100 Hz) – during intense brain activity

(cognitive function, memory & learning)

2. **Lambda waves:** in occipital area during visual attention

➤ Clinical significance of EEG:

1. Localization of brain tumors
2. D.D. of grand mal & petit mal epilepsy
3. Diagnosis of stage of sleep & sleep disorders
4. Confirmation of brain death

3. Physiology of Sleep

➤ Definition:

It is a physiological state of temporary unconsciousness from which the person can be aroused by proper stimuli

➤ **Duration:** (in average)

1. Adult: 7 – 8 h/day
2. Newborn: 16 – 18 h/day
3. Old person: 5 – 6 h/day

➤ Waking/Sleeping rhythm:

- Circadian rhythm: Synchronized with the daily light/dark cycle (How?)
- Physical & psychological factors affect the onset & duration of sleep, e.g.:
 1. Cold & fear → prevent its onset
 2. Fatigue & boredom → facilitate its onset

3. In days preceding examinations → students may use sleep as a legitimate escape way to avert studying

➤ Sleep cycles & types of sleep:

- Sleep cycles = 4 – 5 cycles/night

- Each cycle lasts $\approx$ 100 min. & is composed of 2 types of sleep:

(a) Slow wave sleep = non-REM sleep

(b) Paradoxical sleep = REM sleep

- Sleep starts normally by slow wave sleep (for $\approx$ 80 min.), then REM sleep follows (for $\approx$ 20 min.)

- This is repeated cyclically with gradual prolongation of the REM sleep periods

	Slow wave (non-REM) sleep	Paradoxical (REM) sleep
1. Timing:	At the start of sleep	After the 4 th stage of non-REM sleep
2. Duration:	80% / sleep time	20% / sleep time
3. Depth of sleep & EEG findings:	4 stages: * <u>Stage 1 (very light sleep):</u> Theta waves * <u>Stage 2 (light sleep):</u> Sleep spindles (bouts of large alpha-like waves) that superimpose on theta waves * <u>Stage 3 (moderately-deep sleep):</u> Delta waves (3 Hz) * <u>Stage 4 (deep sleep):</u> Delta waves (1 Hz)	<u>Light sleep:</u> Small rapid irregular beta waves (like an alert person)
4. Eyes:	Deviate up + Miosis	Rapid eye movements

5. Dreams:	Absent {if present → not remembered (Why?)}	Present & can be remembered
6. Erection:	Absent	Frequently occurs
7. HR, RR, BP:	↓↓	↑↑
8. Ms. tone:	Slight ↓	Marked ↓↓↓ (Sleep paralysis)
9. Sleep talking & walking:	Present	Absent
10. Awakening:	Variable (according to the stage)	Difficult (high threshold)

➤ Mechanisms (theories) of sleep:

Depression of CC 2ry to (–) of ARAS activity; by either:

(1) Passive mechanism:

Passive (–) of ARAS activity, due to its fatigue or elimination of its exciting stimuli

(2) Active mechanism: (the real mechanism)

Active (–) of ARAS activity, due to (+) of sleep centers:

	Non-REM sleep	REM sleep	Waking/sleeping rhythm
Center:	Raphe nuclei (in upper MO & lower pons)	Nuclei of RF (in pons)	Suprachiasmatic nucleus (in ant. hypoth.) → Synchronize waking/sleeping rhythm with 24 h light/dark cycle
Ch. tr.:	Serotonin	ACh	Serotonin

➤ Physiological changes during sleep:

1. CVS:

HR & BP: ↓ during non-REM sleep & ↑ during REM sleep

2. Respiratory system:

RR: ↓ during non-REM sleep & ↑ during REM sleep

3. Nervous system:

(a) Voluntary activity & sensory perception: abolished

(b) Reflexes:

- Most reflexes: disappear
- Flexion-withdrawal reflex: depressed, but can still be elicited (?)
- Plantar reflex: +ve Babinski sign

4. Ms. tone, MR & body temp.: ↓↓

5. Endocrinal secretions:

(a) Most hormones: ↓↓

(b) GH & ADH: ↑↑

(c) ACTH & cortisol: ↓↓ to minimum at midnight & ↑↑ to maximum at 8 am

➤ Sleep disorders:

1. **Insomnia:** (the commonest)

*Definition: Lack of sleep in spite of convenient surrounding environment

*Causes: Psychological (fear or anxiety) – Analeptic substances – Irritative lesions of post. hypoth. – Damage of sleep centers

*Effects: ↓↓ Concentration & memory – Bizarre behavior (temporary neurosis & psychosis) – Hallucinations

2. **Hypersomnia:** Prolonged sleep

3. **Narcolepsy:** Irresistible desire to sleep during daytime

4. Non-REM sleep may be associated with:

- (a) Sleep-talking
- (b) Sleep-walking (somnambulism)
- (c) Bed wetting (nocturnal enuresis)

5. REM sleep may be associated with:

- (a) Excessive swallowing
- (b) Teeth grinding (bruxism)
- (c) REM behavior disorder: no ↓ in ms. tone

6. Sleep apnea

Lecture (16): Functions of the different parts of the eye (I)

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Eye Ball

➤ Structure of the eye ball:

(A) 3 layers:

1. Outer fibrous protective layer:

- a) Sclera: post. 5/6, opaque
- b) Cornea: ant. 1/6, transparent

2. Middle vascular layer (Uveal tract):

- a) Choroid: post. 5/6, pigmented, contains BVs
- b) Ciliary body: ant. ring, contains ciliary ms., ciliary processes & suspensory ligaments (SL)